North Ridge Highway Intersection Upgrade

Preliminary pavement design memorandum

1. Pavement design inputs

The pavement draft assumes 2026 AADT 7,200 vehicles/day, 10% trucks and 1.0% annual non-compounded growth. The preliminary subgrade input is CBR 5%. These traffic assumptions have not been reconciled with NRH-TIA-001.

Trial structure: 110 mm asphalt concrete over 250 mm granular base. This is an option for review, not a verified structural design. Final design must apply the Alberta Pavement Design Manual with applicable Design Bulletin 77 amendments and accepted project inputs.

2. Traffic coordination calculation

Draft base-year truck count = 7,200 × 0.10 = 720/day. Over 20 years from 2027 through 2046, sum of year offsets 1 through 20 = 210. Two-way truck-pass screen = 720 × (20 + 0.01 × 210) × 365 = 5,807,880. At 50% directional split this is 2,903,940 truck passes.

Using the TIA inputs on the same screening basis gives 1,092 × (20 + 0.02 × 210) × 365 = 9,645,636 two-way truck passes, or 4,822,818 at 50% directional split. These are not ESALs. No axle-load factors, truck-class distribution or lane allocation has been applied.

Project	Issue information
NRH-26104	2026-09-19
North Ridge Industrial Lands Partnership	Issued for review
Prepared / checked	JL / AK

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3. Required resolution

The differing 7,200 versus 7,800 AADT and 10% versus 14% truck shares lack a documented source reconciliation. Growth methods may legitimately differ by design method, but any difference must be explained and supported.

Do not approve the trial pavement section from these truck-pass calculations. Obtain accepted base-year data, design life, classification, load factors and design-lane ESALs; confirm subgrade and drainage; then compare structural alternatives under the applicable guidance.